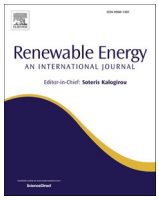
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Effects of hydrogen bond acceptor with benzyl group in deep eutectic solvents on pretreatment performance of bamboo residue

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ABSTRACT

Nine benzylammonium chloride-based deep eutectic solvents (DESs) were prepared and used to disrupt the natural recalcitrance of bamboo cell walls, to simultaneously produce fermentable sugars and lignin by-products. Under pretreatment conditions of 130 °C and 6 h, nearly 90 % of lignin was removed during the tributyl benzyl ammonium chloride/oxalic acid (TBBAC/OA) pretreatment of *Dendrocalamus giganteus Munro* bamboo (DG), accompanied by a satisfactory recovery of cellulose (~84 %). The enzymatic digestibility of TBBAC/OA-DG released 83.72 % of fermentable sugars, which is about twice as high as tetrabutylammonium chloride/oxalic acid pretreated DG (TBAC/OA-DG) without benzyl groups as the hydrogen bond acceptor in the DES system. The structural characteristics of hydrogen bond acceptors (HBA) in DESs remarkably influenced the removal of lignin and xylan, which resulted in a loose structure after DES pretreatment, then increased the enzymatic accessibility of bamboo. Furthermore, the obtained regenerated lignin has a narrow molecular weight distribution and excellent solubility in organic solvents, which demonstrated potential industrial applications. These findings not only explain the relationship between HBA structure and bamboo fractionation, but also provide new insights for the selection and design of DES in biomass pretreatment.

1. Introduction

Bamboo has fast growth cycle, high production rate and holocellulose content, and is expected to become a promising substitute for wood resources during the biorefinery processing [1]. *Dendrocalamus giganteus Munro* (*D. giganteus*) bamboo extensively distributed in southwest of China were often used in the production of construction and other daily commodities. It is estimated that the utilization of these bamboo is only about 30 %, while the annual processing waste is as high as 20 million tons [2,3]. The production of value-added compounds (sugar-based compounds and bioethanol) from bamboo residues have attracted more attention for the development of bamboo biorefinery [4–7]. However, due to the high degree of lignification, high vascular bundles density, and complex cell wall structure of bamboo, efficient enzymatic saccharification has always been one of the main obstacles and expensive processes [8,9]. Therefore, biomass pretreatment has become an essential step towards high conversion of polysaccharide fraction as it can break the recalcitrance of biomass, and improve its accessibility to cellulase.

The green pretreatment technology of biomass has always been a research hotspot for the sustainable development of biorefinery industry. In addition to achieving ideal pretreatment efficiency, it requires

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the use of fewer toxic and harmful chemicals, the generation of fewer fermentation inhibitors, and the maximum biomass utilization [10]. During the developed pretreatment technologies, deep eutectic solvents (DESs) have become a promising method due to its low price, easy preparation, low toxicity and high biodegradability [11]. DESs are formed through mixing hydrogen bond donors (HBD) and hydrogen bond acceptors (HBA) in a specific molar ratio under heating conditions. HBD and HBA self-associated by means of hydrogen bond interactions, forming a homogeneous eutectic mixture with a melting point much lower than that of its individual component [12,13]. DES pretreatment has been proven to be effective in disrupting the cell wall structure of biomass, fractionating lignocellulose and increasing enzyme accessibility to cellulose substrates. There have been many reports on the effects of HBDs (such as organic acids, amino acids, and sugars) on DES pretreatment of biomass, and it has been found that the design of HBDs was highly correlated with the selective extraction of lignin from lignocellulosic biomass [14]. However, little attention was paid to whether the HBA structure affected the DES characteristics and biomass pretreatment.

Hydrogen bond acceptors (HBAs) play a crucial role in the formation of DES and the number of hydrogen bonds in the system. Choline chloride (ChCl) is commonly used as HBAs in the synthesis of DES, because it was highly biodegradable, non-toxic, and can form DES with a wider range of HBDs [15,16]. Recently, benzylammonium chloride as an HBA has received increasing attention in DES pretreatment of biomass, because it has the same positively charged ammonium moiety and chloride ions as ChCl [17–21]. It belongs to the aromatic quaternary ammonium salt and has better phase transfer catalytic ability compared to ChCl, which meant that the solubility of lignin in benzylammonium chloride-based DES may be stronger than in ChCl-based DES. Corn cob was pretreated with benzyltrimethylammonium chloride-lactic acid, removing 80.8 % of xylan and 63.4 % of lignin [17]. Xie et al. [19] found that DES pretreatment of benzyltrimethylammonium chloride/formic acid (BTMAC/FA) can remove 78.03 % of pine lignin at 140 °C and 2 h. Bamboo shoot shells using benzyl triethylammonium chloride (BTEAC), formic acid (FA) and water pretreatment were performed on enzymatic hydrolysis, 87.0 % of monosaccharides was achieved. They found that BTEAC-based DES showed good potential in biomass pretreatment, because compared to ChCl-based DES, it had effective delignification and similar costs [20]. Although DES pretreatment using benzylammonium chloride as HBA has achieved some progress in the fractionation of biomass, efforts are still undertaken to improve its sustainability and diversified development.

In this study, a series of benzylammonium chloride-based DES with oxalic acid as an HBD were prepared; their properties were measured and compared. Then the benzylammonium chloride-based DESs were applied to treat bamboo residues for subsequent bioconversion of polysaccharides and preparation of lignin by-products. The relationship between the structure of benzylammonium chloride and the pretreatment performance of DES was investigated. The physicochemical properties of DES pretreated samples were systematically studied, and the mechanism of DES pretreatment with different HBAs was explored. Finally, the regenerated lignin was further characterized to promote the DESs-based biorefinery industrialization.

2. Materials and methods

2.1. Materials and reagents

The *Dendrocalamus giganteus Munro* bamboo (*D. giganteus*, *DG*), were obtained from Baoshan City (Yunnan Province, China). They were ground and screened to 40–60 mesh for pretreatment experiments. Cellulase CTec2 (200 FPU/mL) was provided by Sigma Aldrich in Shanghai (China). Oxalic acid (OA, ≥99.5 %), trimethyl benzyl ammonium chloride (TMBAC, ≥98 %), triethyl benzyl ammonium chloride (TEBAC, ≥98 %), tripropyl benzyl ammonium chloride (TPBAC, ≥98 %), tributyl benzyl ammonium chloride (TBBAC, ≥98 %), tetraethyl ammonium chloride (TEAC, ≥98 %), tributyl ammonium chlorine (TBAC, ≥98 %), choline chloride (ChCl, ≥98 %), tetradecyl dimethyl benzyl ammonium chloride (TDBAC, ≥98 %), benzyl dimethyl (2-hydroxyethyl) ammonium chloride ([BzCh]Cl, ≥98 %), were purchased from Shanghai Macklin Biochemical Co., Ltd.

2.2. DES pretreatment using different HBAs and the extraction of regenerated lignin

Benzylammonium chloride and oxalic acid (OA) in a molar ratio of 1:1 was mixed using a mild heating stirring method until a clear liquid was obtained. Due to the high viscosity of TPBAC/OA-DES and its inability to stir, 15 % water was added to adjust its viscosity to approximately the same as TEBAC/OA-DES in this study. The Kamlet-Taft polarity parameters (π*, β and α) of DES were measured using dye indicator (N, N-dimethyl-4-nitroaniline, 4-nitroaniline and Nile red) [18,22].

The prepared DES was mixed with bamboo powder in a high temperature reactor at a ratio of solid and liquid (1:10) and 130 °C for 6 h. Then the obtained mixture was subjected to vacuum filtration; the solid residue was repeatedly washed with hot water. The washing solution and the pretreatment solution was mixed together; this liquid fraction was treated with a 1:3 (v/v) ethanol/water mixture as antisolvent and then evaporated in a rotary evaporator to obtain DES regenerated lignin. The DES pretreated bamboo was defined based on the name of HBD and different HBAs. For example, *D. giganteus* treated with tetraethylammonium chloride (TEAC) and oxalic acid (OA) was designated as TEAC/OA-DG. The chemical compositions of untreated and treated DG were measured using the National Renewable Energy Laboratory (NREL) protocol [22]. Three repeated measurements were conducted on all samples. The delignification, hemicellulose loss, and cellulose recovery of DG were determined according to the content of each component in the pretreated substrates, as described in our previous research [23–25].

2.3. Enzymatic saccharification of pretreated bamboo

Enzymatic hydrolysis of untreated and treated DG was performed in sodium citrate buffer solution (50 mM, pH 4.8) in 50 mL enzymatic bottles at 5 % (w/v) substrate loading. The mixtures were incubated at 50 °C in a shaking incubator with 150 rpm for 72 h, and the dosage of cellulase CTec2 was 20 FPU/g-sample. The supernatant was withdrawn periodically and the amount of releasing reducing glucose was determined using HPLC (Agilent 1260, USA), with an Aminex HPX-87H column and 5 mM H_2SO_4 (flow rate of 0.6 mL/min) as the mobile phase. Another experiment, Tween 80 (150 mg/g substrate) was added for reducing the nonproducing adsorption between pretreated samples and cellulase, then further improving the glucose yield during the hydrolysis process. All hydrolytic reactions were performed in duplicates. Glucose yield from enzymatic hydrolysis of pretreated DG was calculated based on following formula:

$$\text{Glucose yield } \% = \frac{\text{Glucose in the supernatant}}{\text{initial Glucan in bamboo} \times 1.11} \times 100\% \quad (1)$$

2.4. Enzymatic adsorption isotherms on treated D. giganteus

The enzymatic adsorption was performed according to the method described in our previous reports [22–24]. The bamboo powder (2 %, w/v) was incubated with a range of cellulase C2730 (0.05–1.0 mg/mL) at 4 °C and 150 rpm for 3 h. After reaching equilibrium, the separation of solid-liquid phase was carried out at 10000 rpm for 10 min, then the protein amount of the supernatants was determined by Bradford assay. Two replicate determinations were run for all samples. Nonlinear regression was performed on the adsorption data using Langmuir

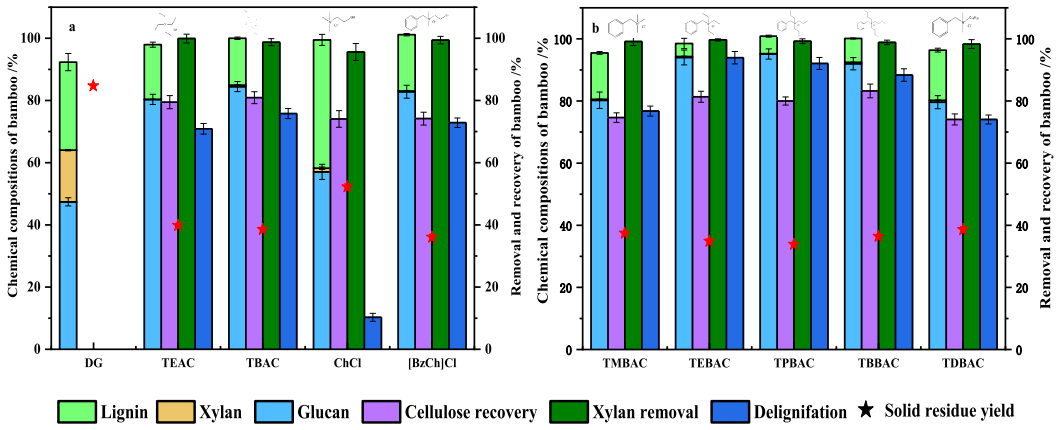


Fig. 1. The effect of different hydrogen bond acceptors in the DES pretreatment on the chemical composition of *D. giganteus*, (a) HBA without benzyl group, (b) HBA containing benzyl group.

isotherms ($\Gamma = \Gamma_m KC/(1+KC)$), where C represented the content of free cellulase in the solution (mg/mL). The adsorption parameters were determined, including Langmuir constant (K), maximum adsorption capacity (Γ_m), and distribution coefficient ($R = \Gamma_m \times K$, L/g) [24,25].

2.5. Analytical methods

2.5.1. Characterization of pretreated samples

The hydrophobicity and cellulose accessibility of bamboo after the DES pretreatments can be obtained based on the adsorption methods of Rose Bengal and Direct Red. Polymerization degree of different samples was determined using the cupriethylene diamine method. Two replicate determinations were run for all samples. A Sigma 300 (Germany) scanning electron microscope (SEM) was used to analyze the surface morphology of samples. X-ray photoelectron spectroscopy (XPS) was obtained on an Escalab 250Xi 201 spectrometer (Thermo Fisher Scientific Inc., US), and the elemental compositions and surface lignin concentration was estimated as previously described [25]. The crystallinity indexes (CrI) of substrates were determined with X-ray diffractometer (Ultima IV, Rigaku, Japan) from 5° to 40° at 8°/min of scanning speed, and was estimated according to the method of Thygesen et al. [26]. The total specific surface areas of pretreated sample were carried out on Micrometritics ASAP 2020 analyzer (USA).

2.5.2. Characterization of DES-regenerated lignins

The M_n (number average) and M_w (weight average) molecular weights of DES-lignins were investigated using Gel permeation chromatography (GPC, Agilent 1260 Infinity II, USA). Detailed structural changes in lignin were acquired by NMR spectrometer (Bruker AVIII 600, Germany); the sample preparation and operation methods were based on previous studies [27].

2.6. Computational analysis

All HBA, HBD and DES molecules were geometrically optimized at B3LPY/6-31+G** level using Gaussian16 software [28]. Then a single-point energy calculation was carried out at B3LYP/def2-TZVPP level, which was used to evaluate hydrogen-bond interaction energy between HBAs and HBDs by Formula 2. After single-point energy calculation, molecular wave function information was obtained and Multiwfn program was used to calculate the Molecular Polarity Index (MPI) of the DESs molecules by Formula 3 [29,30]. The MPI values characterized the polarity of the DESs system. Molecular surface electrostatic potential contour maps were drawn from the electrostatic potential information, which can be derived from wave function information calculated before with Formula 4 [31,32].

$$E_{H\text{-}bond} = E_{DES} - E_{HBD} - E_{HBA} \quad (2)$$

where the $E_{H\text{-}bond}$ refers to hydrogen-bond interaction energy, the E_{DES} interaction energy of DES molecules, the E_{HBD} interaction energy of HBA, and the E_{HBA} interaction energy of HBD.

$$MPI = \frac{1}{A}\iint_S |V(r)|dS \quad (3)$$

where A represents the molecular surface area, V the molecular electrostatic potential at any spatial point r, and the integration is performed over the molecular surface S.

$$V(r) = \sum_A \frac{Z_A}{|R_A - r|} - \int \frac{\rho(r')dr'}{|r' - r|} \quad (4)$$

where Z_A is the potential of positive charge on nucleus A to point r, R_A is the position of the nucleus A and ρ(r') is the molecule's potential caused by electron density of point r'.

3. Results and discussion

3.1. Effect of benzylammonium chloride-based DES on component separation of bamboo biomass

A series of DESs were synthesized using nine ammonium chloride and oxalic acid (OA) to investigate the effect of HBAs on the pretreatment performance of *D. giganteus*. The changes in chemical composition of DG after these DES pretreatments were shown in Fig. 1. As seen in Fig. 1, raw DG comprised 47.42 % cellulose, 16.61 % xylan, and 28.31 % lignin. All tested DES pretreatments resulted in a decrease of solid yield, indicating the dissolution of some chemical components in bamboo. Among the nine DESs, ChCl/OA provided the highest solid recovery (61.57 %), meant that its separation effect of bamboo cell wall components was not as good as other DES pretreatments. All DES pretreatments, including ChCl/OA, exhibited significant removal of xylan, over 96 %, mainly due to the low stability of hemicellulose during acidic DES pretreatment under high temperature conditions. The free protons (H^+) in acidic environment can catalyze and cleave ester and ether bonds between lignin and xylan, thereby separating and extracting amorphous structures of hemicellulose [33].

It was found that there was a clear correlation between the lignin removal and the structures of HBAs in DES system. As for DES pretreatments with benzylammonium chloride as HBA, including TMBAC/OA, TEBAC/OA, TPBAC/OA, TBBAC/OA, TDBAC/OA and [BzCh]Cl/OA, the delignification was 76.76 %, 83.06 %, 92.08 %, 88.36 %, 74.06

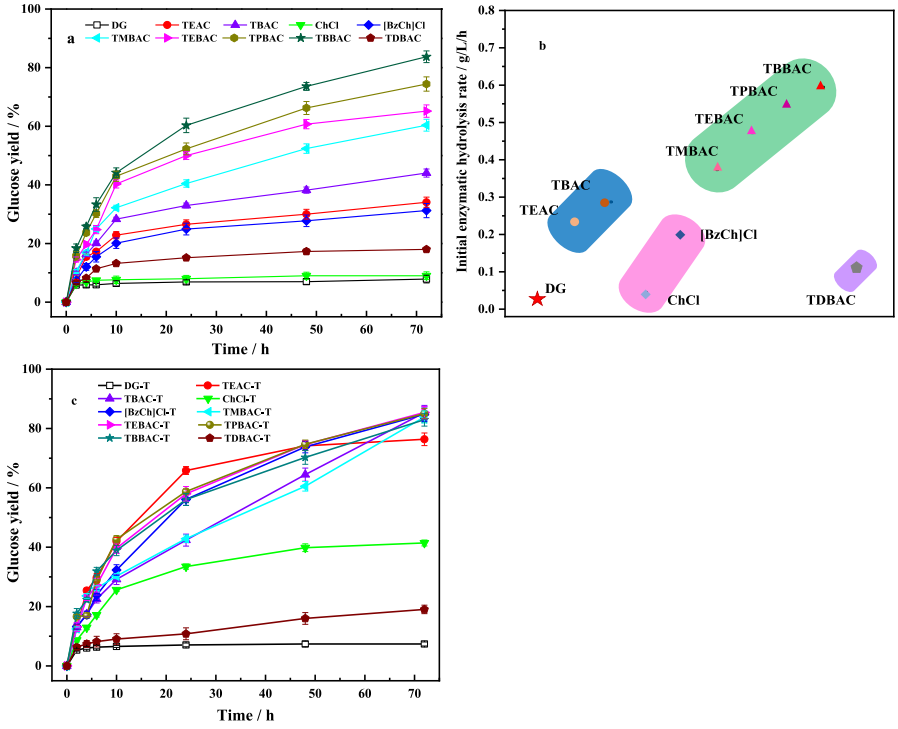


Fig. 2. The effect of DES pretreatment with different hydrogen bond acceptors on the enzymatic hydrolysis efficiency of bamboo, (a) 72 h glucose yield, (b) initial enzymatic digestibility rate, (c) 72 h glucose yield with addition of Tween80.

% and 72.85 %, respectively, meant that the native recalcitrance of bamboo was significantly reduced. This was attributed to the strong hydrogen bond formed between the chloride ions in HBA and the hydroxyl groups in the solute, which is beneficial for the solubilization of lignin [34,35]. When the length of the alkyl chain increased, the delignification of DES pretreatment shows a trend of first increase and then gradually decrease, with TPBAC/OA and TBBAC/OA pretreatment having the higher delignification (92.08 % and 88.36 %). It was interesting that although the carbon chain structure of TBBAC was longer than that of TPBAC, the removal of lignin by TBBAC/OA was slightly lower than that of TPBAC/OA. Subsequently, TDBAC were also selected as HBA to analyze the effect of longer carbon chain on the delignification. It was observed no significant difference in the removal of lignin (76.76 % of TMBAC and 74.06 % of TDBAC), suggesting that the stronger steric hindrance to some extent hindered the removal of lignin. Meanwhile, compared with TMBAC and TDBAC, these HBAs (such as ChCl, TEAC and TBAC) lacked benzyl groups in their structures. The delignification decreased to 10.29 %, 70.89 %, and 75.79 %, respectively, implying that the benzyl structure of HBA could play an important role during the DES pretreatment of bamboo biomass. In comparison, a relatively poor delignification ability (10.29 %) was observed from the pretreatment of ChCl/OA. The structure of choline chloride contains hydroxyl groups, while [BzCh]Cl as HBA contained both benzyl and hydroxyl groups. It was observed that the delignification of [BzCh]Cl/OA increased to 72.85 %, which is about 7 times higher than that of ChCl/OA, suggesting that benzyl structure of HBAs demonstrated great effects on bamboo deconstruction. It was obvious that the HBA containing only benzyl group in the DESs had the best dissolving effect on lignin, followed by HBAs with both benzyl and hydroxyl groups, and HBA with only hydroxyl groups has poor delignification. In addition, it can be seen that lignin and xylan were dissolved during the DES pretreatment process, while the solubility of these DES to cellulose was poor. The pretreated residue had a cellulose retention ratio of 74.06–83.78 %. The large number of intermolecular and intramolecular hydrogen bonds in the cellulose structure made it difficult to dissolve in DES.

As is well known, the delignification of DESs with quaternary ammonium cations as HBA depended on the ability of negative halogen anions to form lignin-chloride intermediates, which can affect the proton catalyzed depolymerization of lignin [14]. All tested HBAs are ammonium chloride, and its chloride ions can form strong hydrogen bonds with lignin, promoting the stretching and breaking of LCC structure and resulting in the dissolution of lignin. However, HBAs containing benzyl groups exhibited stronger ability of lignin removal, while the presence of hydroxyl groups in HBA had a negative effect on delignification. TMBAC, TEBAC, TPBAC, TBBAC, TDBAC and [BzCh]Cl belong to quaternary ammonium salts with aromatic ring, and their phase transfer catalytic ability are stronger than that of non-aromatic quaternary ammonium salts, which can improve the lignin solubility in the DESs [18]. Combined with the effect of free protons (H^+) in acidic DES, the depolymerization of lignin was accelerated, effectively separating lignin in bamboo biomass. Benzyltrimethylammonium chloride/formic acid removed 78.03 % of pine lignin at 140 °C for 2 h [19]. Triethylbenzylammonium chloride (TEBAC) is an aromatic ring quaternary ammonium salt. Liu et al. [18] found that when wheat straw (WS) was treated with DES of TEBAC/lactic acid, 79.73 % of lignin was removed, and 91.27 % of hydrolysis yield was achieved using cellulase of 35 FPU and β-glucosidase of 82 CBU.

3.2. Enzymatic saccharification of the pretreated bamboo residues

After treating DG with DESs, the obtained solid residues were hydrolyzed with enzyme of 20 FPU/g sample. As shown in Fig. 2, the 72 h hydrolysis yield of raw DG were only 7.86 %. By comparison, it can be found that the enzymatic saccharification efficiency of bamboo wasters can be significantly enhanced after the benzylammonium chloride-based DES pretreatment. Among them, the TBBAC/OA pretreatment brought the higher glucose yield (83.72 %), while TPBAC/OA, TEBAC/OA, and TMBAC/OA pretreated DG gave 74.42 %, 65.18 % and 60.36 % glucose yield, respectively. It suggested that with the extension of HBA

carbon chain in the benzylammonium chloride-based DES, the enzymatic hydrolysis yield of pretreated substrates showed an increasing trend. The glucose yield of TBBAC/OA-DG increased by 38.7 % compared to that of TMBAC/OA. This phenomenon may be due to the decrease of lignin content in TBBAC/OA-DG. As previously stated, the formation of strong hydrogen bonds between chloride ions and lignin, as well as the presence of aromatic ring structures in HBAs, contributed to the significant removal of lignin, which increased the exposure of carbohydrate in the pretreated substrates and improved the subsequent enzymatic hydrolysis yield. Moreover, it was found that glucose production was not entirely correlated with the above delignification. Although TPBAC/OA pretreatment resulted in the highest delignification (92.08 %), the hydrolysis yield was a little lower than that of TBBAC/OA. This was mainly because that except for the content of lignin, there were other factors that affected the enzymatic saccharification yield. It meant that the structure of TPBAC/OA-DG, such as the crystallinity, degree of polymerization, specific surface area, and hydrophilic/hydrophobic properties, is not beneficial for its enzymatic saccharification. A lower glucose yield of TDBAC (17.99 %) was also observed, which is caused by the excessively long carbon chain of HBA due to its stronger steric hindrance.

As shown in Fig. 2a, the enzymatic hydrolysis efficiency of DG pretreated with benzylammonium chloride-based DES was significantly better than that of substrates lacking benzyl in HBAs. The cellulose conversion rate of TBAC/OA pretreated DG was 44.04 %, followed by TEAC/OA (34.08 %) and ChCl/OA (9 %) pretreatments, which were also correlated with the carbon chain length of HBA to a certain extent. The enzymatic hydrolysis yield of [BzCh]Cl/OA improved by about 4-folds than that of ChCl/OA, due to the introduction of benzyl structure. Compared with a non-aromatic ring HBA, benzylammonium chloride-based DES pretreatment showed a higher delignification (88.36 %), and correspondingly, the hydrolysis yield of pretreated DG increased by 2–4 times. The similar correlation between the benzyl structure of HBAs and the pretreatment performance was also observed in the initial enzymatic digestibility rate (Fig. 2b). The initial rate (within 6 h) of four pretreated substrates (TMBAC/OA-DG, TEBAC/OA-DG, TPBAC/OA-DG and TBBAC/OA-DG) were 0.37, 0.47, 0.55 and 0.59 g/L/h, respectively, which were about 2 times higher than TEAC/OA -DG and TBAC/OA-DG. The initial rate values of ChCl/OA-DG (0.044 g/L/h) were lower than that of [BzCh]Cl/OA-DG (0.199 g/L/h), indicating that the presence of benzyl groups in the HBA structure was beneficial for bamboo pretreatment. The higher initial rates in benzylammonium chloride-based DES pretreated samples may be related to the lower lignin content and substrate characteristics. The DES pretreatment containing benzyl groups in HBA has been used in the pretreatment process of various biomass, and the satisfactory performance has been achieved compared to DES pretreatment at the absence of benzyl groups in HBA. Zhang and coworkers have demonstrated that the DES of benzyltriethylammonium chloride (BTEAC) and formic acid (FA) can deconstruct the eucalyptus chips within 2 h at 130 °C, obtaining pure cellulose fibers [36]. Benzyltrimethylammonium chloride (BTMAC)/lactic acid (LA) pretreatment can remove 80.8 % of xylan and 63.4 % of lignin under 140 °C and 2 h, leading to 94.0 % of sugar yield under the enzyme usage of 15 FPU/g sample [17]. The DES of benzyltriethylammonium chloride and thymol has a better efficiency in extracting food dyes than the DES of choline chloride and thymol [37]. This was mainly because aromatic quaternary ammonium salts have stronger phase transfer capability, which can be dissolved in water and organic solvents. The negative and positive ions in quaternary ammonium salts formed an ion pair in the aqueous phase, which can transfer negative ions from the aqueous phase to the organic phase, while in the organic phase, negative ions have no solventization role [17,18].

The residual lignin and pseudo-lignin of pretreated substrates can inhibit subsequent enzymatic hydrolysis through physical barriers and irreversible adsorption on enzymes. The nonionic surfactant Tween80 were added to enzymatic digestibility to reduce the inhibitory effect of

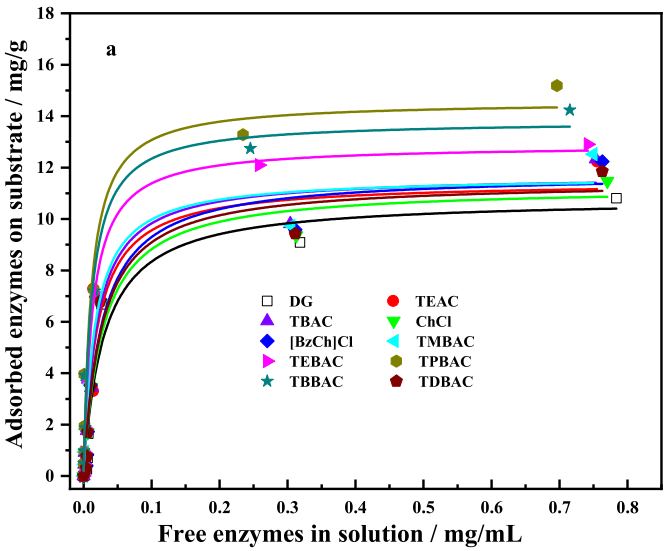


Fig. 3. The effect of DES pretreatment with different hydrogen bond acceptors on the enzymatic adsorption performance.

lignin on enzyme by forming lignin-Tween80 complexes [38,39] and the results are shown in Fig. 2c. As expected, the addition of Tween80 increased the glucose release of pretreated samples with high residual lignin content. Tween 80 slightly improved the 72 h saccharification yields of TBBAC/OA-DG from 83.71 % to 85.9 %, due to the less lignin in TBBAC/OA-DG. However, after adding Tween80, the 72 h glucose yields of the TMBAC/OA-DG, TEBAC/OA-DG, TPBAC/OA-DG was remarkably increased to 75 %, 85.4 % and 85.02 % respectively, improved by about 25–31 % compared with the enzymatic hydrolysis without Tween80. The addition of Tween80 also exhibited enhanced effect on enzymatic saccharification of TEAC/OA-DG, TMAB/OA-DG and ChCl/OA-DG. Non-productive adsorption between enzymes and lignin was an important factor hindering the enzymatic saccharification. The above comparison confirmed that cellulase interacts with Tween80 to form the Tween80-enzyme complexes, which improved the electrostatic repulsion interaction between substrate lignin and enzyme, and reduced the non-productive binding of enzyme on the substrate [40,41]. It is worth noting that although the lignin content of TDBAC/OA-DG was 16.15 %, there was no obvious positive effect of the Tween80 addition on the enzymatic digestibility. This is mainly because TEBAC itself is a cationic surfactant that can adsorb on the surface of the pretreated substrate and clean these lignin droplets during the pretreatment process.

By comparing the enzymatic saccharification efficiency of bamboo treated with different pretreatment methods (see supplementary material), it can be concluded that, except for the bamboo treated with p-TsOH and NaOH, DG pretreated with TBBAC/OA maintained high glucose production in 14 tested bamboo samples. Furthermore, compared with DG pretreated with TBBAC/OA, the enzymatic hydrolysis efficiency of bamboo treated with p-TsOH and NaOH combination was improved by an additional 20 IU/gsubstrate of β-glucosidase. Deep eutectic solvents (DESs) are of great significance for the development of bamboo biorefinery due to their low cost, ease of preparation, low toxicity, and higher bioconversion rate. The TBBAC/OA pretreated DG achieved the satisfactory enzymatic hydrolysis yield and ethanol production, which will reduce the energy consumption of the pretreatment process and the cost of enzymatic saccharification, thus realizing the economic feasibility of bioethanol production. The DES technology with HBA containing benzyl group for bamboo fractionation may provide a new method for bamboo pretreatment and provide some guidance for bamboo biorefinery.

Table 1
Langmuir adsorption isotherm parameters of cellulase adsorption on pretreated bamboo.

Pretreated Substrates	Γ_m (mg/g)	K (mL/mg)	R (L/g)
DG	33.98 ± 1.34	10.79 ± 0.43	0.37 ± 0.01
TMBAC/OA-DG	59.60 ± 1.86	11.65 ± 0.24	0.69 ± 0.032
TEBAC/OA-DG	75.93 ± 0.91	12.88 ± 0.86	0.98 ± 0.012
TPBAC/OA-DG	86.82 ± 1.78	14.57 ± 0.34	1.26 ± 0.054
TBBAC/OA-DG	83.60 ± 1.22	13.82 ± 0.82	1.16 ± 0.024
TDBAC/OA-DG	38.91 ± 0.86	11.46 ± 0.91	0. 44 ± 0.028
TEAC/OA-DG	50.42 ± 1.01	11.45 ± 0.32	0. 58 ± 0.032
TBAC/OA-DG	52.50 ± 1.31	11.70 ± 0.13	0.61 ± 0.012
ChCl/OA-DG	36.26 ± 0.91	11.25 ± 0.42	0.41 ± 0.032
[BzCh]Cl/OA-DG	44.51 ± 1.01	11.64 ± 0.44	0.52 ± 0.039

3.3. Cellulase adsorption isotherms on different DES pretreated D. giganteus

To understand the impact of benzyl groups in HBA structure during the DES pretreatment on the enzymatic saccharification, the adsorption properties between cellulase and substrates were studied based on Langmuir adsorption isotherms (Fig. 3 and Table 1). As seen in Fig. 3, the enzymatic adsorption amount of raw DG was 33.98 mg protein/g-substrate. With the extension of carbon chains in benzyl HBAs structure, the adsorption ability of cellulase on samples was significantly enhanced. The Γ_m values of cellulase on TMBAC/OA-DG, TEBAC/OA-DG, TPBAC/OA-DG and TBBAC/OA-DG were 59.6, 75.93, 86.82 and 83.6 mg/g, respectively, significantly higher than those of raw bamboo. Compared with TMBAC/OA-DG and TEBAC/OA-DG, TPBAC/OA and TBBAC/OA pretreated samples showed the higher cellulase adsorption capacity, which will also be beneficial for their subsequent enzymatic saccharification efficiency. In addition, the TDBAC/OA-DG has fewer binding sites to cellulase enzymes, as TDBAC is itself a cationic surfactant. Meanwhile, there are some differences in the Langmuir constant (K) and distribution coefficient (R) among all DES pretreated samples. The value of K and R improved to 14.57 mL/mg and 1.26 L/g-substrates for TPBAC/OA-DG, and 13.82 mL/mg and 1.16 L/g-substrates for TBBAC/OA-DG, indicating that the adsorption capacity of cellulase on substrates was significantly enhanced. This agreed well with the results of above enzyme hydrolysis, where TPBAC/OA-DG and TBBAC/OA-DG has the higher glucose production (74.42 % and 83.72 %). This was mainly due to DES pretreatment can destroy the complex structure of bamboo by removing most of hemicellulose and part of lignin, resulting in the increase of productive adsorption between cellulase and cellulose substrate [42]. Additionally, the cellulase adsorption activities is related to the physicochemical characteristics of sample. Therefore, in the following experiments, the surface and the structural characteristics of the pretreated samples were determined and compared to explore the differences in their binding strength with enzymes and glucose yield.

3.4. Effect of benzylammonium chloride-based DES pretreatment on physicochemical properties of D. giganteus

The morphology and structure of the untreated and treated bamboo were deeply studied to further evaluate the pretreatment performance, which can provide some information for enhancing enzymatic saccharification yield. The morphology of the pretreated samples is observed by SEM images (Fig. 4). The raw DG exhibited a smooth and compact surface, which can hinder the enzymatic accessibility to cellulose. All DES pretreatments can fracture the recalcitrant structure of bamboo by removing xylan and part of lignin to different extents, which resulted in a rougher and porous surface, and provided a loose substrate for subsequent enzymatic hydrolysis. Furthermore, many folds and small debris appeared on the DES pretreated substrate, providing more adsorption and active sites for cellulase during the enzymatic saccharification. Notably, some spherical substances or droplets appeared on the surface of TEAC/OA-DG, TBAC/OA-DG and ChCl/OA-DG substrates, which is believed to be the lignin migration under severe acidic

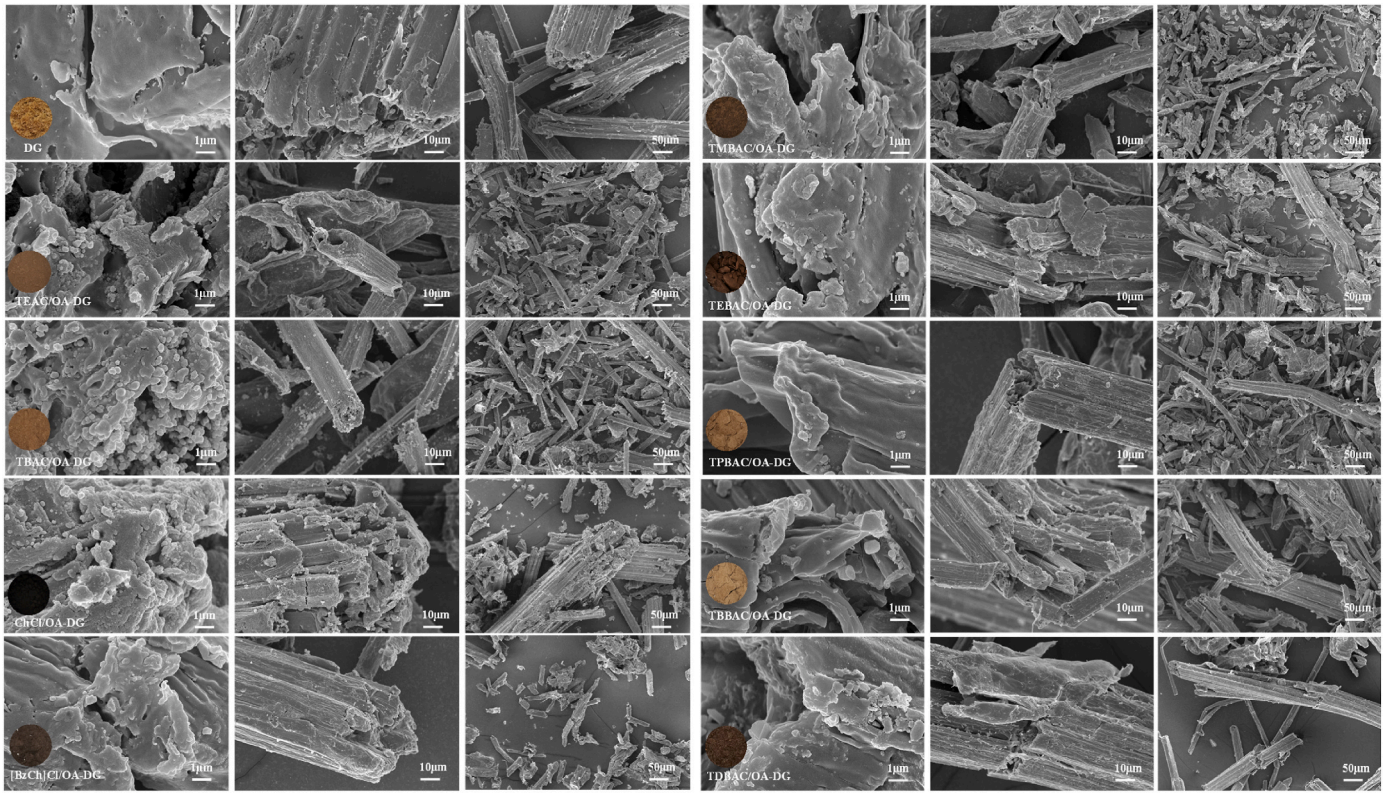


Fig. 4. The surface morphology of pretreated *D.giganteus* with SEM images.

Table 2
Elemental composition and functional groups based on the C1s peaks by XPS analysis.

Sample	O/C ratio	Surface lignin coverage (%)	Surface carbohydrate coverage (%)	Surface chemical bonds (%)			
				C-C, C-H (284.6, eV)	C-O (286.1, eV)	C=O (287.4, eV)	O-C=O (289, eV)
DG	44.72	76.57	23.43	49.44	34.11	6.74	9.72
TEAC/OA-DG	54.31	57.38	42.62	37.02	32.27	17.08	13.63
TBAC/OA-DG	44.65	76.70	23.30	41.56	39.94	11.01	7.48
ChCl/OA-DG	34.86	96.28	3.72	50.46	24.14	12.30	13.10
[BzCh]Cl/OA-DG	49.43	67.13	32.87	43.66	38.12	8.54	9.69
TMBAC/OA-DG	55.93	54.13	45.87	41.89	39.03	9.06	10.01
TEBAC/OA-DG	64.49	37.02	62.98	28.20	42.20	6.45	23.15
TPBAC/OA-DG	61.26	43.47	56.53	40.16	40.84	7.86	11.15
TBBAC/OA-DG	49.77	66.46	33.54	45.09	37.44	8.11	9.36
TDBAC/OA-DG	47.94	72.11	27.89	52.99	34.78	6.56	5.66

Table 3
The molecular weights and different OH groups in DES regenerated lignin.

Lignin	Molecular weights			Aliphatic OH (mmol/g)	Syringyl OH (mmol/g)		Guaiacyl OH (mmol/g)		p-hydroxy phenyl OH (mmol/g)	Total phenolic OH (mmol/g)	S/G ratio (phenolic-OH)
	Mw (g/mol)	Mn (g/mol)	PDI		C	NC	C	NC			
MWL	3484	1357	2.57	3.42	0.05	0.22	0.06	0.7	0.75	1.79	0.36
ChCl/OA-lignin	1432	736	1.95	0.22	0.46	0.50	0.28	0.83	0.71	3.04	0.86
TEBAC/OA-lignin	1186	700	1.69	0.54	0.36	0.87	0.31	0.93	0.83	3.55	0.99
TBBAC/OA-lignin	1745	947	1.84	0.63	0.3	0.79	0.28	0.76	0.71	2.91	1.05

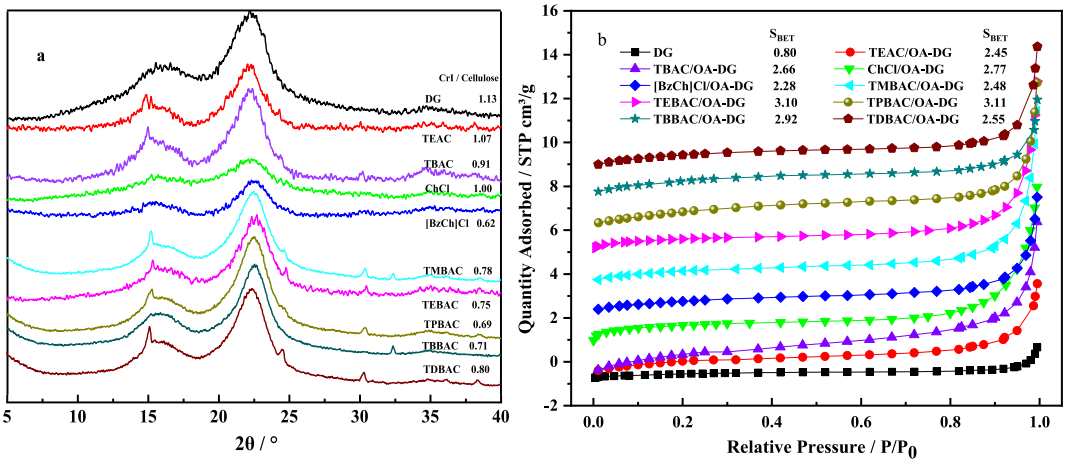


Fig. 5. The effect of DES pretreatment with different hydrogen bond receptors on the crystallinity (a) and BET surface area (b) of *D. giganteus*.

conditions, while lignin droplets were not observed in the SEM images of benzylammonium chloride-based DES pretreated samples. Previous studies have also reported that the Klason (acid-insoluble) lignin content of acid pretreated lignocellulose was higher than that in the raw materials. This was due to the combination of degradation products of hemicellulose and lignin during the severe pretreatment, forming a lignin-like substance [43,44]. The hypothesis of pseudo-lignin forming in TEAC/OA-DG, TBAC/OA-DG and ChCl/OA-DG was further tested through XPS analysis (Table 3). The surface lignin content of ChCl/OA-DG (96.28 %) was much higher than that of raw DG (76.57 %), indicating that the migration of lignin and a small amount of dissolved lignin re-deposited on the surface of substrate. As shown in Table 2, TMBAC/OA, TEBAC/OA, TPBAC/OA, and TBBAC/OA pretreatment overcame the formation and adverse effects of surface lignin. The concentration of surface lignin in the substrates was reduced to 54.13 %, 37.02 %, 43.47 % and 66.46 %, respectively. The C-C/C-H peak intensities of these four substrates are relatively low, at 41.89 %, 28.20 %, 40.16 %, and 45.09 %, respectively. Correspondingly, the amount of C-O/C-O-C increased to 35–40 %, suggesting that the four DES pretreatment methods can reduce the content of condensed lignin and surface lignin, exposing a large amounts of cellulose microfibers, which is consistent with the results of SEM images and chemical composition analysis mentioned above.

The crystallinity index (CrI) of cellulose-rich substrates were studied by X-ray diffraction, and its diffractograms were shown in Fig. 5a. All samples exhibited the typical structure of native cellulose I. The ratio of crystallinity to cellulose content (CrI/cellulose) for raw DG was 1.13, whereas that of the pretreated substrates decreased to 0.62–1.07, which

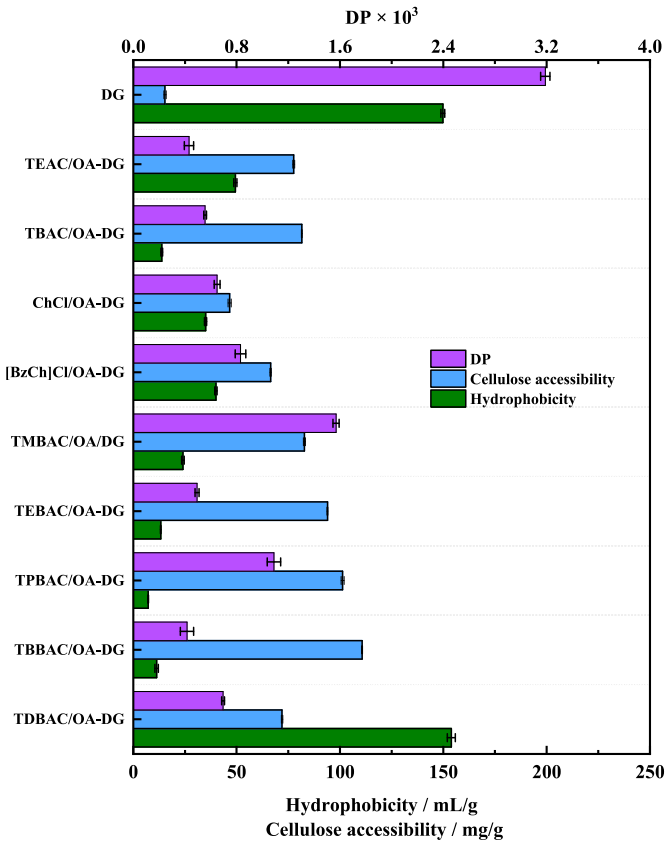


Fig. 6. The physicochemical properties of pretreated *D. giganteus*.

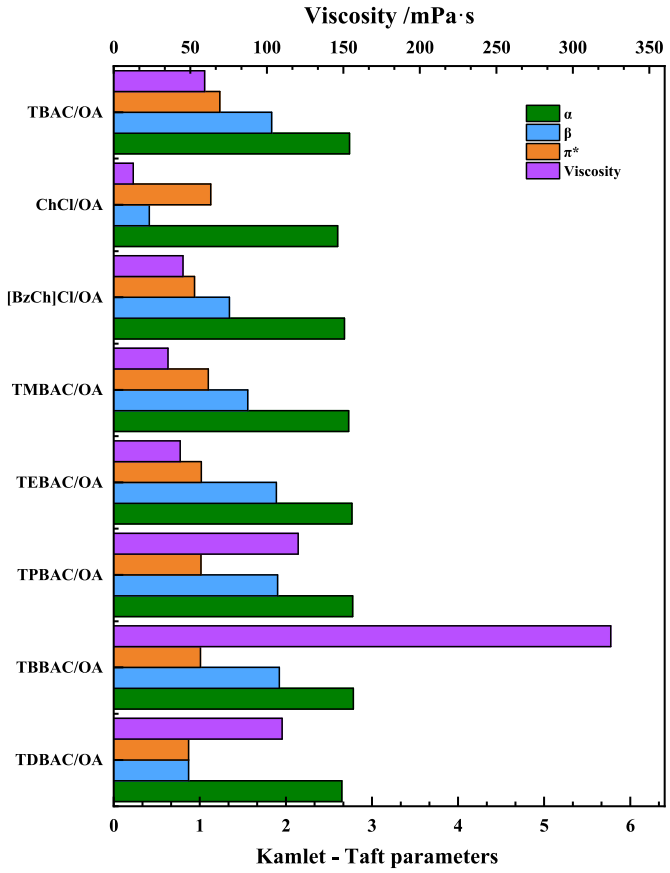


Fig. 7. The Kamlet-Taft empirical parameters of benzylammonium chloride-based DES.

was mainly because of the removals of amorphous xylan and lignin. The CrI/cellulose of the substrates pretreated with TMBAC/OA (0.78), TEBAC/OA (0.75), TPBAC/OA (0.69), and TBBAC/OA (0.71) was lower than that of TEAC/OA (1.07), TBAC/OA (0.91) and ChCl/OA (1.00), which occurred by the hydrogen bonds cleavage in lignocellulosic biomass and the swelling of cellulose components. The results revealed that the benzylammonium chloride-based DES pretreatment was beneficial for destroying the crystal structure of cellulose. The degree of polymerization (DP) can also effectively reflect the hydrolysis difficulty of sample. The lower the DP, the easier the sample is to hydrolyze. Compared with the original DG (3196), DES pretreatment reduced the DP value by 2–7 times (Fig. 6). The TBBAC/OA-DG (416) showed a relative lower DP, with little difference from that of TEAC/OA-DG (431), indicating that more short chain fibers were generated on both types of pretreated substrates compared to the raw materials.

The adsorption characteristics of cellulase are often influenced by the hydrophobicity of the substrate surface. The untreated DG has the higher hydrophobicity of 149 mL/g substrate, followed by the substrates pretreated without benzyl group in HBAs (13–49 mL/g), TPBAC/OA-DG (11.3 mL/g) and TBBAC/OA-DG (7.12 mL/g). This indicated that the hydrophobicity of bamboo residues decreased during the above DES pretreatment, and TPBAC/OA-DG and TBBAC/OA-DG have strong hydrophilicity. The decrease of hydrophobicity reduced the non-productive binding of enzyme to the treated substrates. As shown in Fig. 5b, the BET surface area of bamboo had a great improvement after the DES pretreatment. The BET surface area of raw material was 0.8 m^2/g. Pretreatment with TPBAC/OA and TBBAC/OA resulted in a 3.8-folds increase in surface area relative to DG, with values of 3.12 and 2.92 m^2/g, respectively. This should be attributed to the removal of xylan and lignin, resulting in an increase in substrate porosity. As shown in Fig. 6, compared with others samples, a significant increase in cellulose accessibility was seen in the TPBAC/OA-DG and TBBAC/OA-DG, which indicated that the treatments improved the adsorption ability of substrate on the cellulase. The changes in hydrophilicity and BET specific surface area may be a reasonable explanation for the enhanced cellulose accessibility of TPBAC/OA-DG (101.29 mg/g substrate) and TBBAC/OA -DG (110.73 mg/g substrate). It is well known that there is an obvious positive correlation between cellulose accessibility and enzyme saccharification efficiency [45].

3.5. *Effect of different DES properties on the pretreatment performance*

The Kamlet-Taft solvation discoloration parameters (α, β, and π *) can be used to describe the capacity of DES for providing and accepting hydrogen bonds, as well as the polarizability of solvents and the results were shown in Fig. 7. The HBAs in all tested DES belonged to quaternary ammonium salts and can be divided into two categories: HBA with benzyl group and HBA without benzyl group. As can be shown in Fig. 7, there was no great difference in the α values of all DES systems, as their HBA was oxalic acid and had similar hydrogen donating capabilities. The polarity (π*) and hydrogen bond basicity (β) of DES play an important role in the fractionation of biomass. Higher π* value indicates the studied DES is polarizable solvent. The π * value of DESs containing benzyl group in HBA was significantly lower than that without benzyl group in HBA, indicating that the presence of benzyl group in HBA can enhance the capability of DESs to build non-specific interactions with biomass components. An opposite trend was observed for the change of β values in the five DESs with benzyl group. The TBBAC/OA (1.923) and TPBAC/OA (1.903) exhibited a relatively higher β value than the TMBAC/OA and TEBAC/OA. The Kamlet-Taft parameter β of ionic liquids has an obvious impact on the biomass fractionation efficiency [46, 47]. In addition, as compared with other DES systems, the macromolecular properties of TBBAC and TPBAC typically resulted in stronger hydrogen bond interactions, which also increased the viscosity of

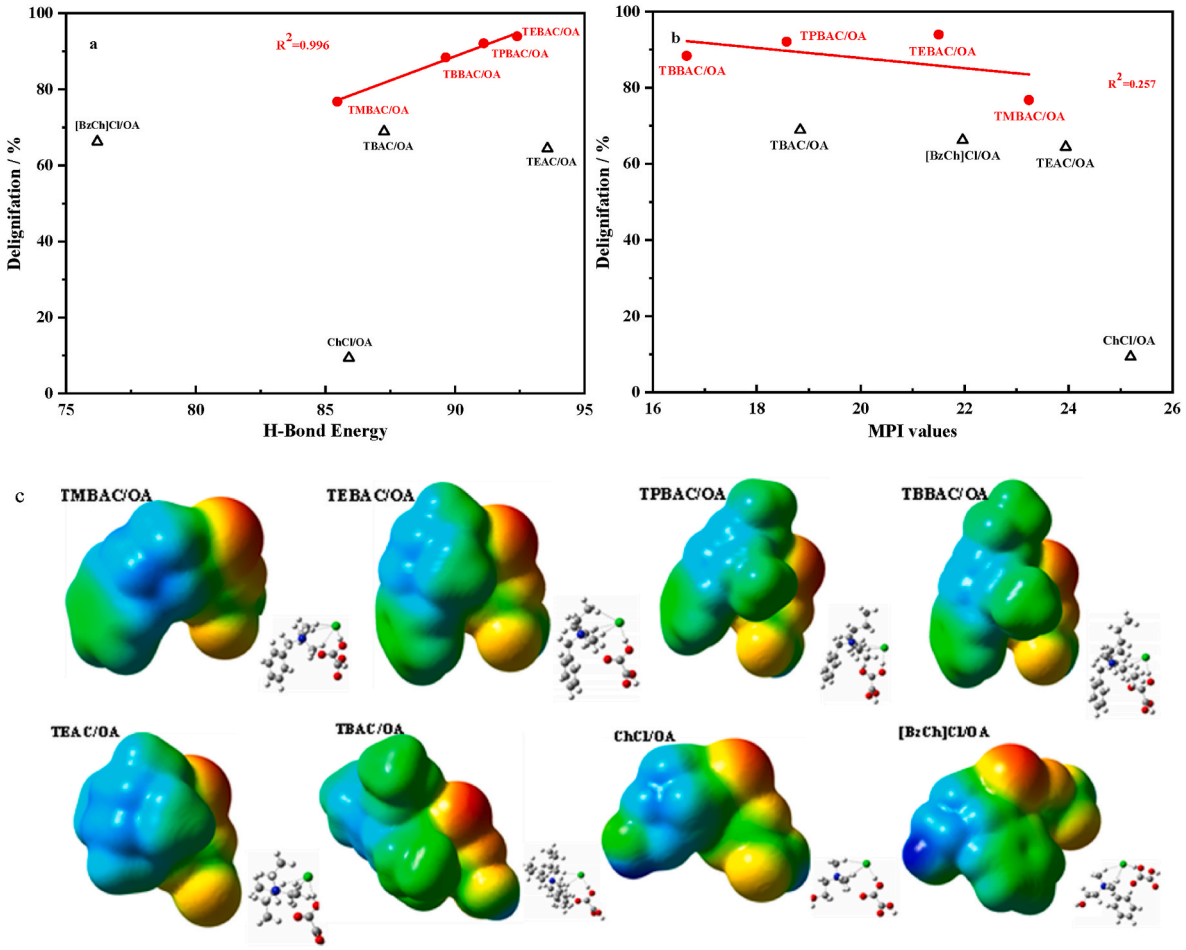


Fig. 8. Scatter plots between delignification (a), H-bond energy (b) and MPI value (c).

TBBAC/OA (324 mPa s) and TPBAC/OA (120 mPa s). The high viscosity of the DES system will affect the mass transfer and fluidity of the substrate during the pretreatment, and the viscosity can be reduced by introducing a third component, such as water, p-TsOH and ethylene glycol [48–50].

The quantum chemical characteristics of the nine tested DESs were related to their delignification during the pretreatment process (Fig. 8). It can be seen that there was no obvious correlation between H-bond energy, molecular polarity index (MPI), and delignification for all DES pretreatments. However, when focusing on the DES pretreatment with benzylammonium chloride as HBA, there was a significant positive correlation (r^2 = 0.996) between the H-bond energy of DES and its delignification (Fig. 8a), indicating that the higher the H-bond energy of DES in this group, the better the delignification. However, the MPI value of this group was not proportional to the delignification. When the MPI value is 21.5, TEBAC/OA has the highest removal of lignin (93.94 %). Compared with the MPI value of DES system, H-bond energy contributed more to the delignification during the biomass pretreatment. Molecular surface electrostatic potential and molecular structure of all tested DES systems were shown in Fig. 8c, where the green part represents non-polar area of DES system. The larger non-polar area has been observed on the benzylammonium chloride-based DES. By comparing TEBAC, TBBAC, and [BzCh]Cl with TEAC, TBAC, and ChCl, a significant increase in delignification was observed, which can be attributed to the increase of non-polar regions on the molecular surface of the DES system with benzyl groups in HBA. Substitution of alkyl groups with benzyl groups in HBA will increase the non-polar portion of the surface, as we can see from the MPI values and molecular surface electrostatic potential maps. Another more important factor for this drastic improvement may be the benzyl group interaction with lignin. The benzene ring of benzylammonium chloride-based DES can interact with the benzene ring of lignin through π-π interactions, which will cause DES molecules to accumulate on the surface of lignin and promote its dissolution in DES. Meanwhile, DES molecules on the surface of lignin may catalyze the breakdown of lignin into fragments.

3.6. *Molecular weight and structural characterization of DES regenerated lignin*

The Molecular weights of raw milled wood lignin (MWL) and the DES regenerated lignin was determined by gel permeation chromatography and listed in Table 3. The weight-average (M_w) of the MWL was 3484 g/mol, whereas the average M_W values of ChCl/OA-lignin, TEBAC/OA-lignin, and TBBAC/OA-lignin were 1432, 1186, and 1745 g/mol, respectively. Compared with MWL, the molecular weights of the DESs lignin reduced by 49–66 %, indicating lignin was dissociated and depolymerized during the DES pretreatment. The M_w of TBBAC/OA-lignin was higher than that of ChCl/OA-lignin, which may be due to the leaching of small molecular weight portion of lignin during TBBAC/OA pretreatment. The polydispersity index (PDI) of three DES regenerated lignin fractions also decreased to 1.95 of ChCl/OA-lignin, 1.69 of TEBAC/OA-lignin and 1.84 of TBBAC/OA-lignin, suggesting that DES pretreatment resulted in narrower molecular weight distribution and these DES lignins became more uniform. It was found that the PDI value of ChCl/OA-lignin was slightly higher than that of TBBAC/OA lignin and TEBAC/OA-lignin, indicated that repolymerization of fragmented lignin units may occur under ChCl/OA pretreatment.

The chemical structures of DES-lignin were explored using 2D HSQC

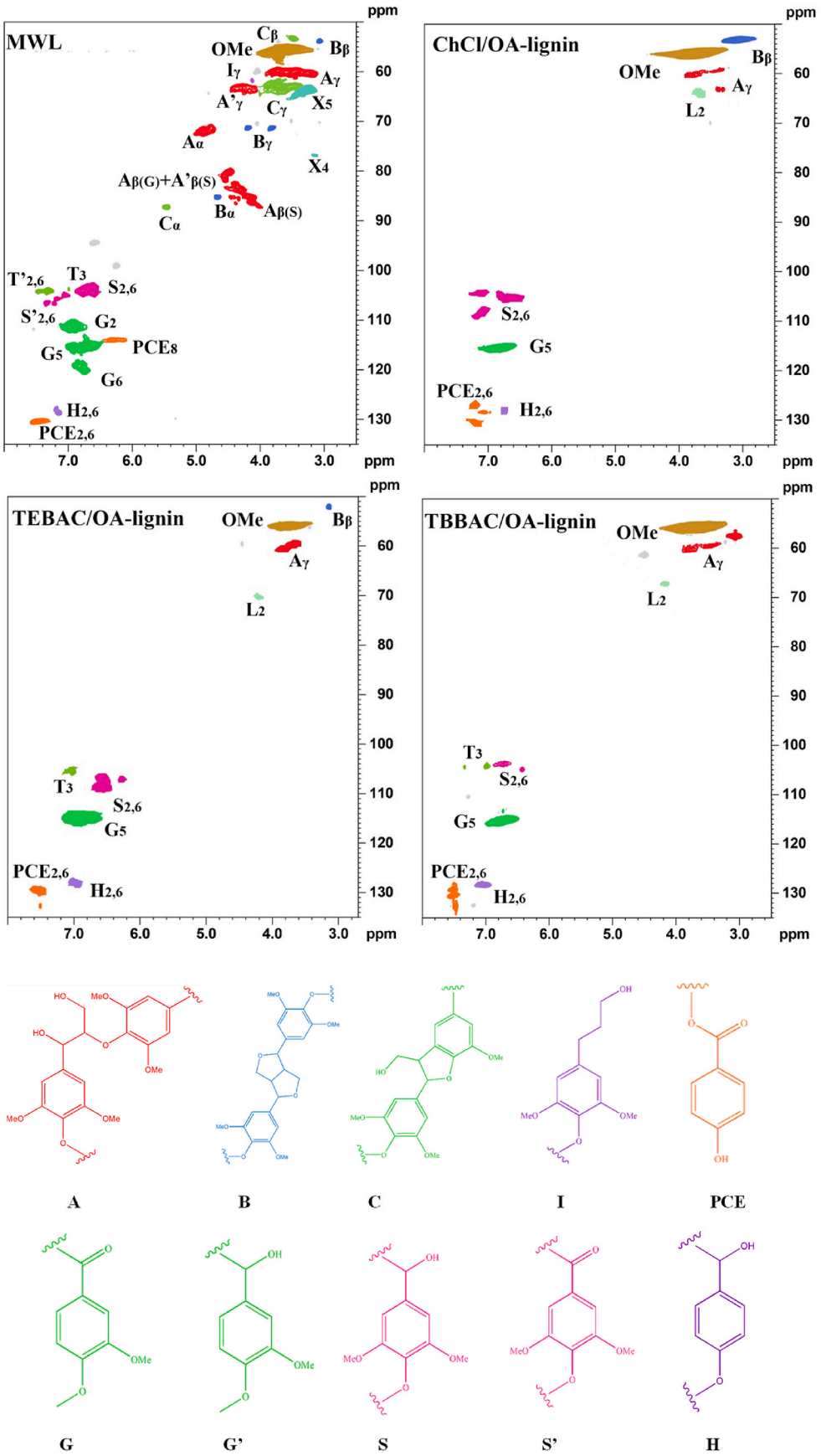


Fig. 9. 2D-HSQC spectra of the MWL and the regenerated lignins obtained with various DES pretreatments.

NMR (Fig. 9). In the side chain region (δ_C/δ_H = 50–90 ppm), for the original bamboo lignin, the obvious signals of the β-5 (C) and β-O-4 (A) linkages were identified, and the weaker signal of the β-β linkages (B) was observed. The methoxyl groups (OCH3) of DES lignin were mostly retained, while the amounts of β-O-4 linkages reduced significantly, indicating that the major aromatic ether bonds of bamboo lignin were mostly cleaved after the pretreatment. Furthermore, the C_γ-H_γ, C_β-H_β and C_α-H_α related signals in the phenylcoumaran (β-5, C) substructure were completely depleted in three EDS lignins. Additionally, the β-β of resinol substructures (B) were also observed in ChCl/OA-lignin, implying that during the ChCl/OA pretreatment process, the cleavage of β-O-4 aryl ether and the condensation of β-β bonds could occur simultaneously. In the aromatic regions (δ_C/δ_H 100.0−150.0/5.90−8.20), the correlated signals of guaiacyl (G), syringyl (S) and p-hydroxyphenyl (H) unit were very obvious in DES lignin samples, suggesting that the extracted lignin was mainly made up of S, G and H unit [51,52].

The ^{31}P NMR spectra of lignin was used to quantitatively determine the active functional groups in the DES regenerated lignins, and the different amounts of OH group in these lignins were summarized in Table 3. For natural MWL, the amount of guaiacyl OH (G-OH) groups were greater than that of syringyl OH (S-OH) groups. This suggested that most of S-type units preferentially participate in the formation of β-O-4 linkage in this lignin, therefore only a small amount of free S-type OH group were found in the ^{31}P NMR spectra. As seen in Table 3, after DES pretreatment, the amount of aliphatic hydroxyl groups decreased, while the amount of total phenolic hydroxyl groups significantly increased, from 0.75 mmol/g to 2.9–3.5 mmol/g. The increase in phenolic OH content was mainly due to the breaking of β-O-4 ether bonds in lignins; The reduction of aliphatic OH was attributed to dehydration and acylation reactions during the pretreatment. In addition, Table 3 also shows a significant increase in the content of non-condensed S-OH and G-OH groups, especially non-condensed S-OH groups. This fact indicated that in the DES system, the β-O-4 ether linkage was very easily dissociated. The increase in the number of S-type and G-OH groups in the condensation indicates that DES pretreatment can lead to the condensation reaction of lignin. Meanwhile, it was observed that the condensed-OH content of ChCl/OA-lignin (0.74 mmol/g) was significantly higher than that of TEBAC/OA-lignin (0.62 mmol/g) and TBBAC/OA-lignin (0.58 mmol/g), which may provide evidence for the high-level occurrence of lignin condensation during ChCl/OA pretreatment. TBBAC/OA pretreatment can significantly hinder the condensation reaction of lignin. Besides, the S/G ratio of lignin can also reflect the changes of structure occur during the processing. In Table 1, the S/G ratio of MWL was 0.36, lower than the values for ChCl/OA-lignin (0.86), TEBAC/OA-lignin (0.99) and TBBAC/OA-lignin (1.05). The increase of S/G ratio was explained by the fact that DES pretreatment was conducive to the preferential dissolution of S units from the cell wall and the easier condensation of G lignin units.

4. Conclusions

This work demonstrated that the benzylammonium chloride-based DES pretreatment can achieve the efficient fractionation and bioconversion of bamboo biomass due to its better delignification and phase transfer catalytic ability. The removal of xylan and lignin was significantly improved to 96.86 % and 88.36 % under the optimal conditions with TBBAC/OA system at 130 °C for 6 h. Meanwhile, 83.72 % of enzymatic efficiency has been achieved by TBBAC/OA pretreatment, increased by 2–4 times than DES pretreatment with non-aromatic ring HBA. Moreover, DES-lignin had several unique features, including high purity, narrower molecular weight distribution and good solubility in many organic solvents. Therefore, this study provides a potential method for green and efficient fractionation and saccharification of bamboo cell wall components.

CRediT authorship contribution statement

Zhaoming Liu: Writing – original draft, Resources, Methodology, Investigation. **Yongzhi Fu:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis. **Shulin Gao:** Software, Methodology, Data curation. **Zhengjun Shi:** Visualization, Supervision, Resources, Conceptualization. **Dawei Wang:** Visualization, Validation, Supervision. **Mizi Fan:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Taohong Li:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition. **Jing Yang:** Writing – review & editing, Writing – original draft, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: All authors had declared no financial interests and personal relationships.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2025.122891.

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